EX NO: 14

RAW SOCKETS TO IMPLEMENT PACKET SNIFFING.

AIM:

Write a Code using RAW sockets to implement Packet sniffing.

PROGRAM:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_call back (packet):
    if IP in packet:
        ip_layer = packet[IP]
        Protocal = ip_layer.proto
        src_ip = ip_layer.src
        dest_ip = ip_layer.dest

        protocal_name = " "
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"

        print(f" Protocol : {protocol_name}")
        print (f " Source IP : {src_ip} ")
        print (f"Destination IP : {dst_ip}")
        print ("_" *50)

sniff (iface = 'wi-fi", prn=packet_call back, filter = "ip", store=0
```

OUTPUT:

```
C:\Users\JAIHARISH\ python packet.py.

Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6
- - - - - - - - - - -
Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.141.117
- - - - - - - - - -
Protocol: TCP
Source IP: 192.168.1.6
Destination IP: 57.155.146.117
- - - - - - - - - -
```

RESULT:

Thus, a code using RAW sockets to implement Packet sniffing was written.